IoT Based Smart Distribution Board for Real-Time Monitoring, Smart Appliances Control and Intelligent Appliances Scheduling

Hafiz Luqman Riaz[1], Munib Irfan[1], Muhammad Shahroz[1]
[1]Department of Electrical Engineering, The Islamia University of Bahawalpur, Pakistan
Corresponding author: Dr. Abdur Raheem, abdur.raheem@iub.edu.pk

1. ABSTRACT

The world energy situation is experiencing an important revolution due to the increasing electricity demand, prevalence of electrical appliances, and the pressing necessity to use energy effectively. Traditional electrical distribution boards, a fundamental constituent of power delivery systems, are mostly passive equipment that can only do simple power protection and routing. This paper will introduce an IoT based Smart Distribution Board (SDB) that integrates the concept of intelligence, flexibility and autonomy into one system, redefining the traditional distribution systems. The suggested system allows tracking electrical parameters in real-time, smart control of the appliances used, and automatic planning of use depending on the preferences of the users, the presence of power, and the priorities of its use. In contrast to other market solutions that provide these capabilities as independent or cloud-based solutions, the proposed SDB implements all the capabilities in the distribution board itself but can be made work in the offline or low-connectivity mode. The innovation can be applied especially in areas where load shedding, grid instability, and limited digital infrastructure are a problem and it is meant to enhance efficiency in energy consumption, reliability, lower their operational expenses, and close the technological divide between urban and rural power consumers.

Keywords: Smart Distribution Board, Internet of Things, Intelligent Energy Management, Load Scheduling, Smart Power Systems

2. Product Description

The Smart Distribution Board (SDB) presented in this paper will be an intelligent variant of substituting the traditional electrical panels with an IoT-enabled framework. It combines sensing, computation, decision-making, and control in one enclosure and is easy to use by the electricians as well as end users. The system uses sophisticated voltage and current sensors with high accuracy to obtain real-time electrical data on every circuit connected to it. The results of these measurements are handled by an embedded control unit that has the capacity to run prescribed logic, adaptive algorithms and user defined schedules. Smart switching elements make it possible to selectively and flexibly control individual or grouped loads. One of the main distinctive characteristics of the system is the hybrid operational architecture: advanced monitoring and analytics can use the internet connection, but the

SDB is not based on uninterrupted connectivity. All the control logic and routines used in scheduling are stored in memory locally, so the functionality can continue even when the network is unavailable. The system can also be coupled with renewable sources like solar photovoltaic systems that allow prioritization of the dynamic loads in situations where there is a shortage of energy supply. The design is robust, flexible and appropriate in different deployment contexts.

3. Introduction

Power is a significant asset that contributes to the current life, industry and development of nations, whereas in most developing countries the system of distribution of power is out-of-date and unstable. Nations such as Pakistan also suffer load shedding, voltage fluctuation and occurrence of unpredictable blackouts, and the old distribution boards cannot solve such problems because they do not provide any idea on consumption patterns and active management of loads. New products in IoT and embedded systems have facilitated smart energy solutions, however, numerous commercial systems only monitor or remote switch, usually involving multiple devices, constant cloud connectivity and expensive. The paper suggests a comprehensive Smart Distribution Board (SDB), which will incorporate real time monitoring, intelligent control, and automated scheduling in one unit. The SDB is a viable, cost-efficient platform, which can enhance efficiency, reliability, and flexibility especially in under-served or infrastructural limited communities.

4. Flow Chart

The functional architecture of the proposed system provides a smooth interaction between the power electronics, sensing modules and intelligent control logic. The grid, solar or backup electrical power passes through integrated sensor smart circuit breakers. Measurement of real time electrical parameters are constant and relayed to an embedded controller where these parameters are analyzed with decision making algorithms. The data is compared in terms of pre-defined thresholds and schedules and priorities, because of which appliances are automatically managed by the system, and safety and efficiency are optimized. Active feedback to local and remote user inputs forms a process of self-correction, converting the distribution board into an active energy management system.

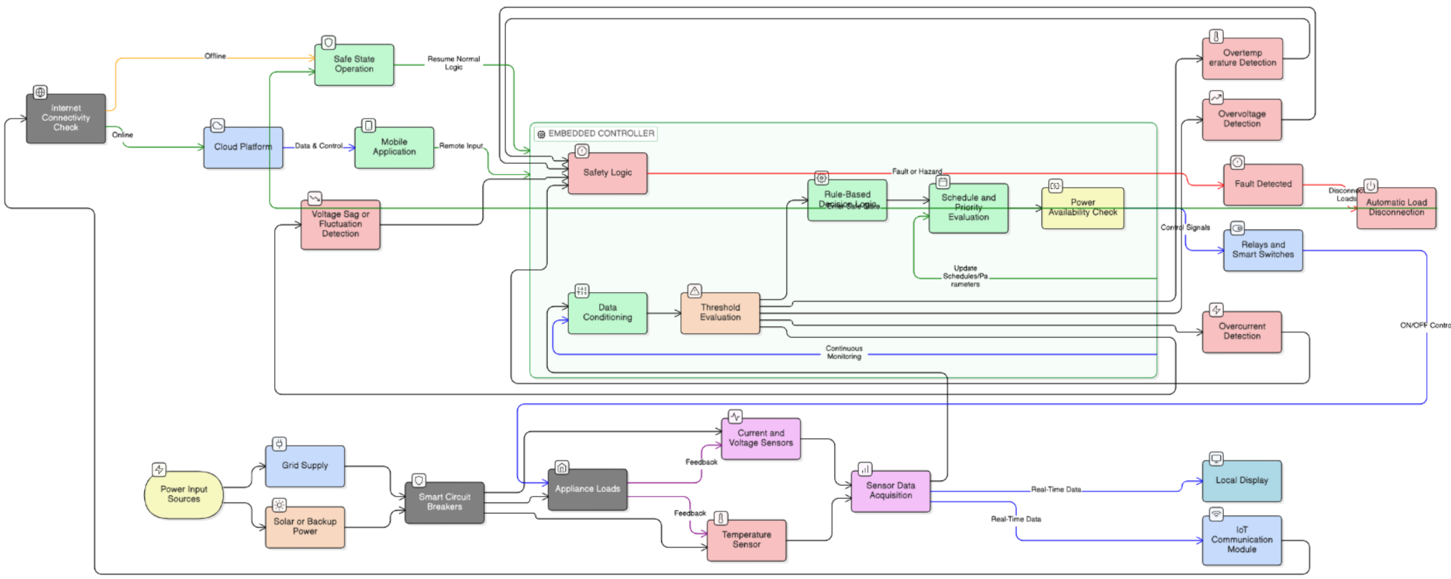

Figure 1. This flowchart is a smart energy management system that is based on IoT. The first step includes the sensors that this device measures and then integrates these electrical parameters into data and then transmits the data to the cloud to process it. The system allows monitor in real time and intelligently schedule based on the analyzed information. Lastly, there are control commands which are created, run on devices and made better through user feedback and refinements, to maximize energy uses.

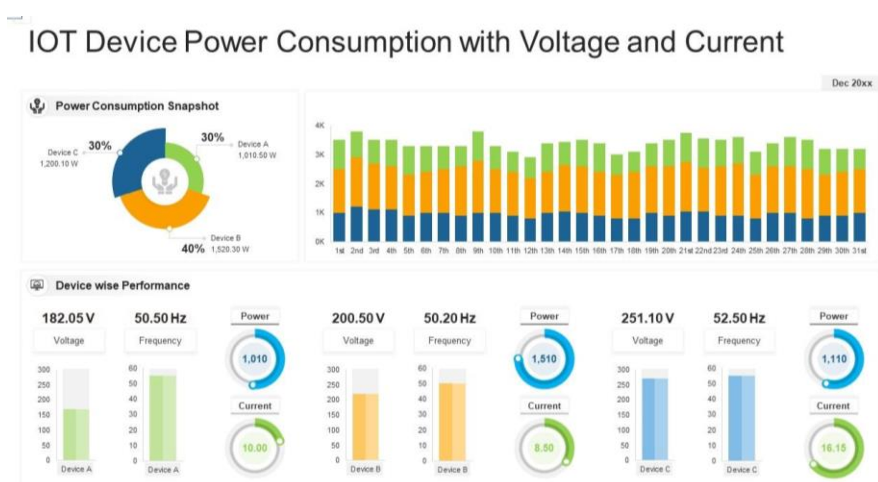

Figure 2. This dashboard shows the detailed overview of energy consumption in 3 units of IoTs in that Device B uses the largest portion of power consumption at 40% (1,520.30 W) of

the full power load. The stacked bar chart is used to visualize the predictable power consumption during the month, and the granular performance section will be used to monitor important electrical characteristics such as Voltage, Frequency, and Current. Interestingly, the current draw of Device C is the highest at 16.15 A and when compared to other units allows the operator to recognize the definite load intensity and possible inefficiencies. The dashboard provides real-time monitoring of electrical health and long-term trend prediction of the entire system of devices by keeping this data centralized.

Figure 3. Time series trends of voltage, current, power components, power factor, total harmonic distortion (THD), temperature and vibration.

5. Novelty and uniqueness

The nature of novelty present in the study is the fact that the three most important capabilities (real-time monitoring, intelligent appliance control, and automated scheduling) have been conceptually and functionalized in one smart distribution board. The current solutions in the international market generally spread these features among different devices, software platforms, or cloud-based services. On the other hand, the suggested system represents an autonomous intelligence becoming located on the very edge of the power distribution.

Moreover, this is a major shift in the paradigm of IoT since the system can be used without constant internet connectivity. Local intelligence will make the proposed SDB applicable in rural and industrial applications with unreliable or no connectivity. This contextual scalability, supplemented by scalability and integration of renewable energy, makes it a conceptual and visionary addition to the smart power into infrastructure.

Table 1. Comparison of Traditional DB with IOT based Smart DB

Feature/Capability	***Traditional Distribution Board***	***Proposed IoT-Based Smart Distribution Board***
Real Time Electrical Monitoring	Not available	Available with live parameter visualization
Appliances Level Control	Manual switching only	Intelligent local and remote control
Automatic Load Scheduling	Not available	Fully supported with offline capability
Internet Dependency	Not applicable	Optional (core functions work offline
Energy Consumption Analytics	Not available	Available with daily and historical insights
Load Prioritization During Power Outage	Not supported	Intelligent priority-based load management
Renewable Energy Integration	Limited or none	Seamless integration with solar systems
Suitable For Rural Areas	Basic power distribution only	Designed for low-connectivity environments
Scalability (Residential to Industrial)	Limited	Highly scalable and modular
User Interaction	No user interface	Mobile application and/or local display

6. Benefit to mankind

Smart Distribution Board (SDB) does not just provide technical efficiency benefits to society. It allows users to check and control the usage of energy to create awareness and use it in a responsible manner. Scheduling is automated and does not need unnecessary power consumption, which allows avoiding grid congestion and sustaining energy security. Where load shedding is being done, intelligent load prioritization guarantees the required appliances will not be shut down. In the case of industry, higher reliability and transparency of consumption lead to productivity and minimize losses in operation. Overall, sustainable energy means that integration with renewable sources and reduced carbon-footprints support sustainable energy. Government-wide, the SDB improves the quality of life, promotes economic sustainability and environmental stewardship.

7. Innovation and Entrepreneurial Impact

The Smart Distribution Board (SDB) signifies technological creativity as well as the prospect of the market. It is created to consider viable energy needs and to make its products affordable, expandable, and manufacturable locally, which allows the production, employment, and owning of the technology to native economic regions. The concept of innovation is not only limited to hardware but also on a wider ecosystem of software services, installation, customization, and maintenance. The SDB as an e-disruptive venture has a competitive advantage in serving markets that are not well served by high quality global products and services. Even its flexibility in domestic, commercial and industrial environments intensify its entrepreneurial capacity making it a scalable, powerful tool, which aligns technology with accessibility to the market.

8. Potential commercialization

The commercialization of smart distribution board based on the IoT is high due to growing need of energy efficiency and smart infrastructure. Cost saving and reliability are desired by residential consumers whereas precision, continuity, and optimization are needed by industries. The suggested system can be commercialized in the forms of modules variants that suit the specific load capacity and application areas.

It can also hasten the uptake of the solar energy into the market through strategic alliances with solar energy suppliers, electrical contractors, and infrastructure developers. This is complemented by electrification and sustainability efforts by the government, which increases the viability of commercialization. The proposed SDB has a good potential for successful market penetration by ensuring that technological innovation is matched with economic feasibility.

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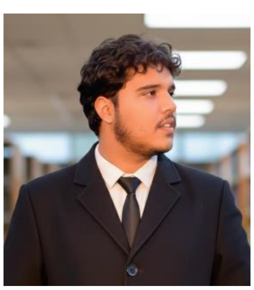
10. Researchers' Biography

	**Hafiz Luqman Riaz** is a final-year Electrical Engineering student at The Islamia University of Bahawalpur, supervised by **Dr. Abdur Raheem**. He specializes in power distribution, protection systems, and IoT-based energy management, with expertise in real-time monitoring and smart automation. His technical background includes hardware prototyping, microcontroller programming, and practical experience with solar photovoltaic systems.
	**Muhammad Shahroz Chaudhary** is a final-semester Electronic Engineering student at The Islamia University of Bahawalpur and a researcher at the CARE Lab. He specializes in FPGA-based systems, robotics, and embedded design, with a focus on digital system design and hardware optimization. An MDPI-published author, his work bridges theoretical research with practical, laboratory-validated implementation. He is currently pursuing a career in R&D across the industry and academic sectors.

	**Munib Irfan** is a final-year Electrical Engineering student at The Islamia University of Bahawalpur, supervised **by Dr. Abdul Raheem**. Driven by his experience with rural energy scarcity, he focuses on IoT systems and smart energy infrastructure to address societal challenges. His work emphasizes leveraging intelligent technologies to improve energy accessibility, efficiency, and sustainability through socially impactful engineering solutions.
	**Dr. Abdur Raheem** is an Assistant Professor in the Electrical Power Engineering Department at The Islamia University of Bahawalpur (IUB). Having earned his Bachelor's and master's from UET Lahore and a Ph.D. from University Technology Malaysia (UTM), his research focuses on optimizing renewable energy systems, including wind, bioenergy, and PV arrays. He is an accomplished researcher with numerous publications in international journals, particularly in the fields of electrical engineering and biomass anaerobic digestion optimization.